

Honeywell

JADE™ Economizer Module

(MODEL W7220)

INSTALLATION INSTRUCTIONS



This document describes wiring, power up, basic troubleshooting, and common installation issues for the JADE™ Economizer Module (Model W7220). For further information on mounting and operation, refer to Honeywell document 63-2700.

BEFORE INSTALLATION

Review the following before installing the The W7220:

Electrical

Rated Voltage: 20 to 30 Vac RMS; 50/60 Hz
Transformer: 100 VA maximum system input

Relay Digital Output Rating: 30 Vac

IMPORTANT

All inputs and outputs must be Class 2 wiring.

Inputs

A Mixed air (MA) analog sensor is required on all W7220 units; either an OA (Outdoor Air) sensor for dry bulb change over or an OA Sylkbus sensor for outdoor enthalpy change over is required in addition to the MA sensor. An additional Return Air (RA) Sylkbus sensor can be added to the system for differential enthalpy changeover. JADE can be used with a CO₂ indoor air sensor for Demand Control Ventilation and with a VFD for two speed fan operation.

DCV (CO₂) Sensor (C7232): 2-10 Vdc control signal; minimum impedance >50k ohm.

Outputs

Actuator signal: 2-10 Vdc; minimum actuator impedance is 2k ohm; Sylkbus two-wire output for Honeywell Sylkbus communicating actuators.

Exhaust fan, Y1, Y2 and AUX1 O: Contact closure (24 Vac)

All Relay Outputs (at 30 Vac):
Running: 1.5A maximum
Inrush: 7.5A maximum

When Installing This Product

1. Read these instructions carefully. Failure to follow them could damage the product or cause a hazardous condition.
2. Check ratings given in instructions and on the product to ensure the product is suitable for your application.
3. Installer must be a trained, experienced service technician.
4. After installation is complete, check out product operation as provided in these instructions.
5. For additional detailed information on this product go to customer.honeywell.com and download the product data sheet (lit no 63-2700) or scan QR code located on the label above MAT terminals with your smart phone.

Accessories

50048926-001 2-pin edge connector for sensors (20 pieces per bag)
50048926-002 6-pin edge connector for field wiring (20 pieces per bag)
C7250 20k sensor for MA or OA (dry bulb changeover)
C7400S Sylkbus sensor for enthalpy control in OA and/or RA and RA for differential dry bulb changeover
W7220 PCMOD interface tool for JADE controller and Personal Computer. Go to www.customer.honeywell.com/economizertools for the software
50053060-001 Duct mounting kit for sensors
C7632 or C7232 CO₂ sensors



62-0331-15

INSTALLATION AND SETUP

The Economizer module may be mounted in any orientation. However, mounting in the orientation shown above permits proper viewing of the LCD display and use of the keypad.

NOTE: JADE will be in the “set up” mode for the first 60 minutes after powered. If a sensor for OA air or Sylkbus device (sensor, actuator) is disconnected during the set up mode, the JADE will not alarm that failure. The MA sensor is a system “critical” sensor; if the MA sensor is removed during the set up mode, the JADE will alarm. After 60 minutes the JADE controller will change to operation mode and all components removed or failed will alarm in the operation mode.

WIRING

All wiring must comply with applicable electrical codes and ordinances, or as specified on installation wiring diagrams. Module wiring in the field is terminated to the four screw terminal blocks located on the left and right sides.

Module wiring at the OEM factory is terminated via the header pin terminals located on the left and right sides. The header terminal pins and the terminal blocks have common terminations for the appropriate input or output. The part number for the OEM female mating connector is 0039973997.



WARNING

Electrical Shock Hazard.

Can cause severe injury, death or property damage.

Disconnect power supply before beginning wiring, or making wiring connections, to prevent electrical shock or equipment damage.



CAUTION

Equipment Damage Hazard.

Electrostatic discharge can short equipment circuitry.

Ensure that you are properly grounded before handling the unit.

Each terminal can accommodate the following gauges of wire:

- Single wire – from 18 AWG to 22 AWG solid or stranded
- Multiple wires – up to two 22 AWG stranded
- For the 24 Vac connections: single wire – from 14 to 18 AWG solid or stranded

For S-BUS wiring, the sensors may be mounted up to 200 ft. (61 m) from the JADE controller. When the length of wire is over 100 feet use twisted pair shielded wire.

NOTE: All wiring is polarity insensitive.

Refer to Fig. 1 through Fig. 7 for common wiring configurations. Refer to Fig. 8 for use with the Smart VFD. Figure 8 shows how to make a single-speed blower operate with two speeds with JADE. When the thermostat calls for first stage cooling or independent fan, the VFD will operate the fan at low speed. When the thermostat

calls for first stage heat or second stage cooling, it will operate the fan at high speed. To achieve this, wire relays as shown, and configure the VFD to the minimum frequency M3.3.1 for low speed fan, and for high speed fan configure Preset frequency 1, M3.3.12.

Actuator Wiring Options:

1. The JADE economizer controller can only have one (1) communicating actuator connected to it.
2. Up to four (4) non-communicating and (2) 2-position actuators (1 each on EXH1 and AUX1 O)
3. One (1) communicating and up to four (4) non-communicating and (2) 2-position actuators (1 each on EXH1 and AUX1 O)

When using a 2-position actuator on the AUX1 O the AUX1 O must be programmed for Exh2 and the % open is the % open of the outdoor damper when the 2-pos actuator opens. Connect 24 V to Exh1 and/or AUX1 O and ground to the JADE “C” terminal.

Note the W7220 economizer will not work with an old “black motor” M7415; replace the M7415 with a M7215 or MS3103 DCA.

Table 1. Economizer Module - Left hand terminal blocks.

Label	Type	Description
Top Left Terminal Block		
MAT MAT	20k NTC and COM	Mixed Air Temperature Sensor (polarity insensitive connection)
OAT OAT	20k NTC and COM	Outdoor Air Temperature Sensor (polarity insensitive connection)
S-BUS S-BUS	SYLK Bus	Sylk Bus sensor (polarity insensitive connection)
Bottom Left Terminal Block		
IAQ 2-10	2-10 Vdc	Air Quality Sensor Input (e.g. CO ₂ sensor)
IAQ COM	COM	Air Quality Sensor Common
IAQ 24V	24 Vac	Air Quality Sensor 24 Vac Source
ACT 2-10	2-10 Vdc	Damper Actuator Output (2-10 Vdc)
ACT COM	COM	Damper Actuator Output Common
ACT 24V	24 Vac	Damper Actuator 24 Vac Source

Table 2. Economizer Module - Right hand terminal blocks.

Label	Type	Description
Top Right Terminal Block		
	n/a	The first terminal is not used
AUX2 I	24 Vac IN	Shut Down (SD) or Heat (W) Conventional only or Heat Pump Changeover (O/B) in Heat Pump mode.
OCC	24 Vac IN	Occupied / Unoccupied Input
E-GND	EGND	Earth Ground - System Required
EXH1	24 Vac OUT	Exhaust Fan 1 Output

Table 2. Economizer Module - Right hand terminal blocks. (Continued)

Label	Type	Description
AUX1 O	24 Vac OUT	Programmable: Exhaust fan 2 output or ERV or System Alarm output.
Bottom Right Terminal Block		
Y2-I	24 Vac IN	Y2 in - Cooling Stage 2 Input from space thermostat
Y2-O	24 Vac OUT	Y2 out - Cooling Stage 2 Output to stage 2 mechanical cooling
Y1-I	24 Vac IN	Y1 in - Cooling Stage 1 Input from space thermostat
Y1-O	24 Vac OUT	Y1 out - Cooling Stage 1 Output to stage 1 mechanical cooling
C	COM	24 Vac Common
R	24 Vac	24 Vac Power (Hot)

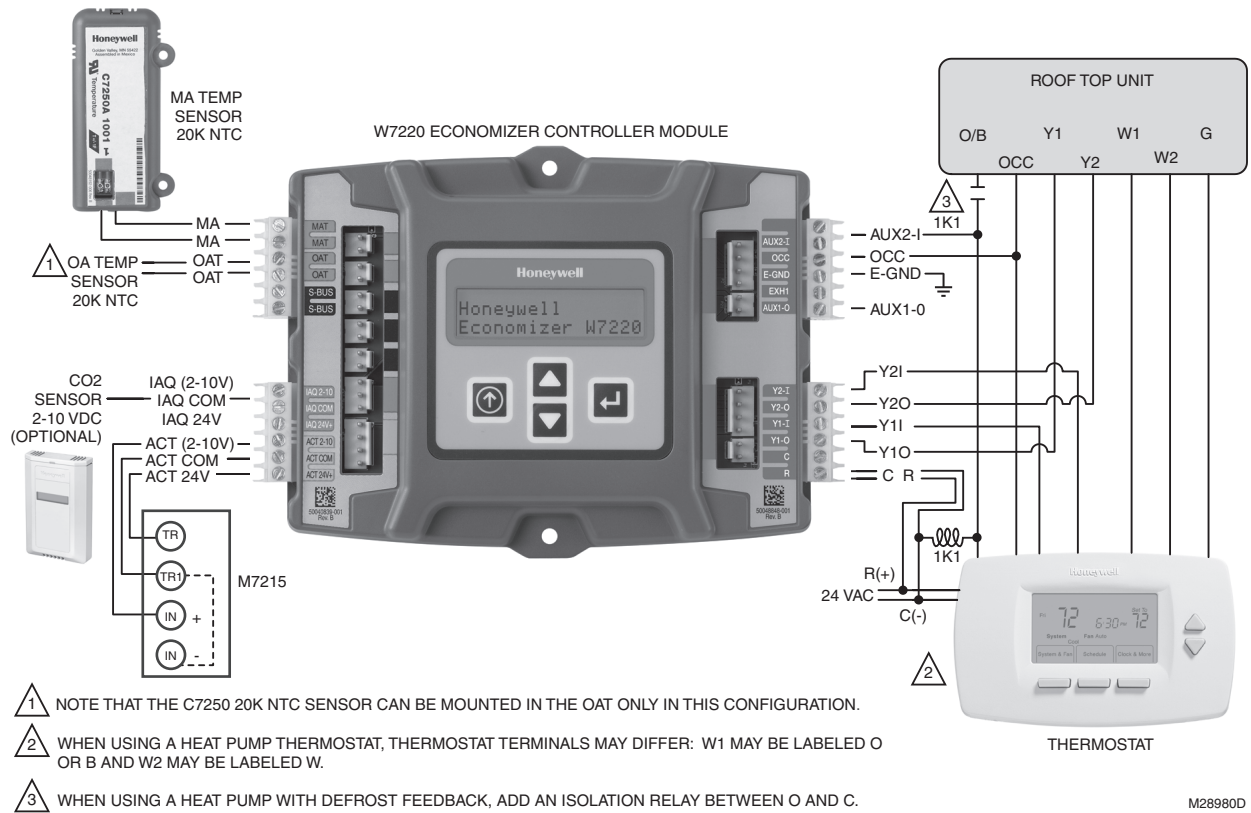


Fig. 1. Stand-alone dry bulb Economizer configuration with black motor M7215.

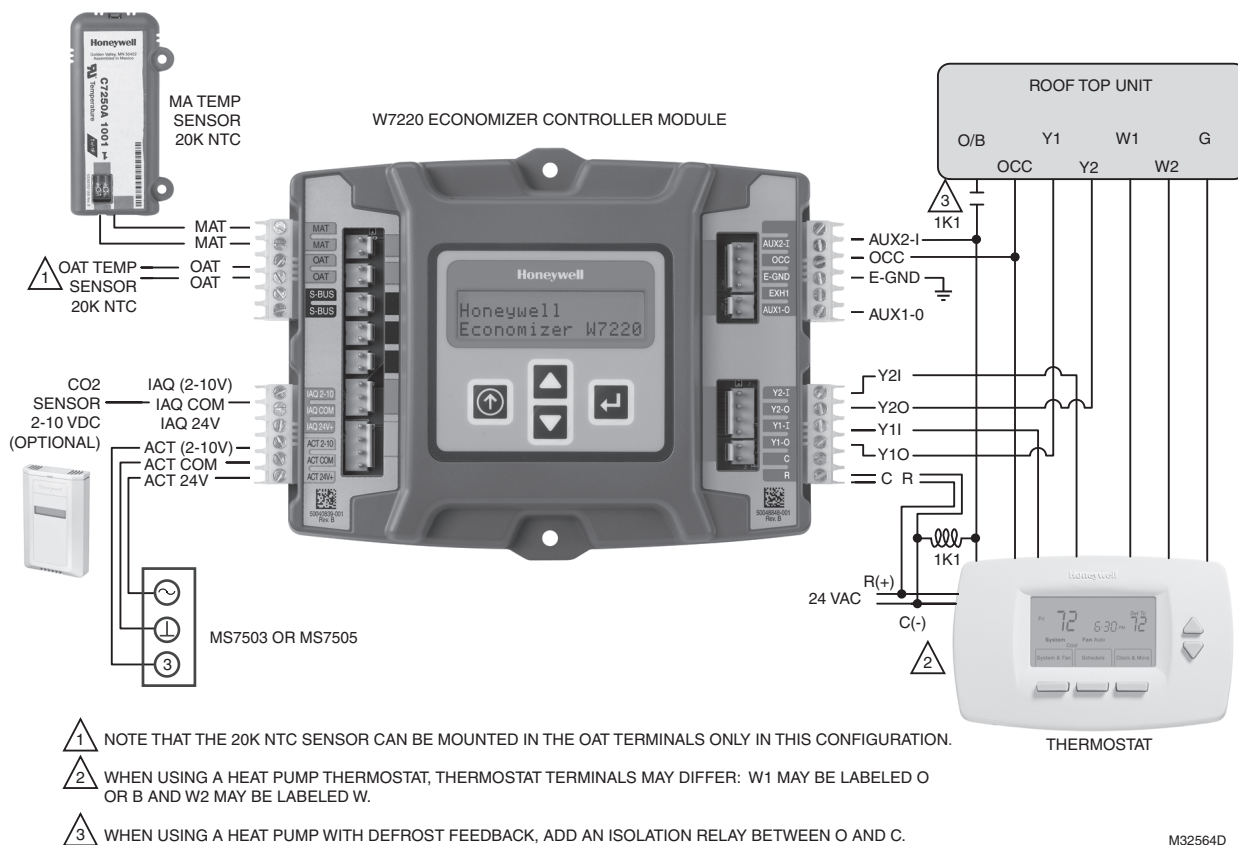


Fig. 2. Stand-alone dry-bulb Economizer configuration with Honeywell MS7503 Direct Coupled Actuator.

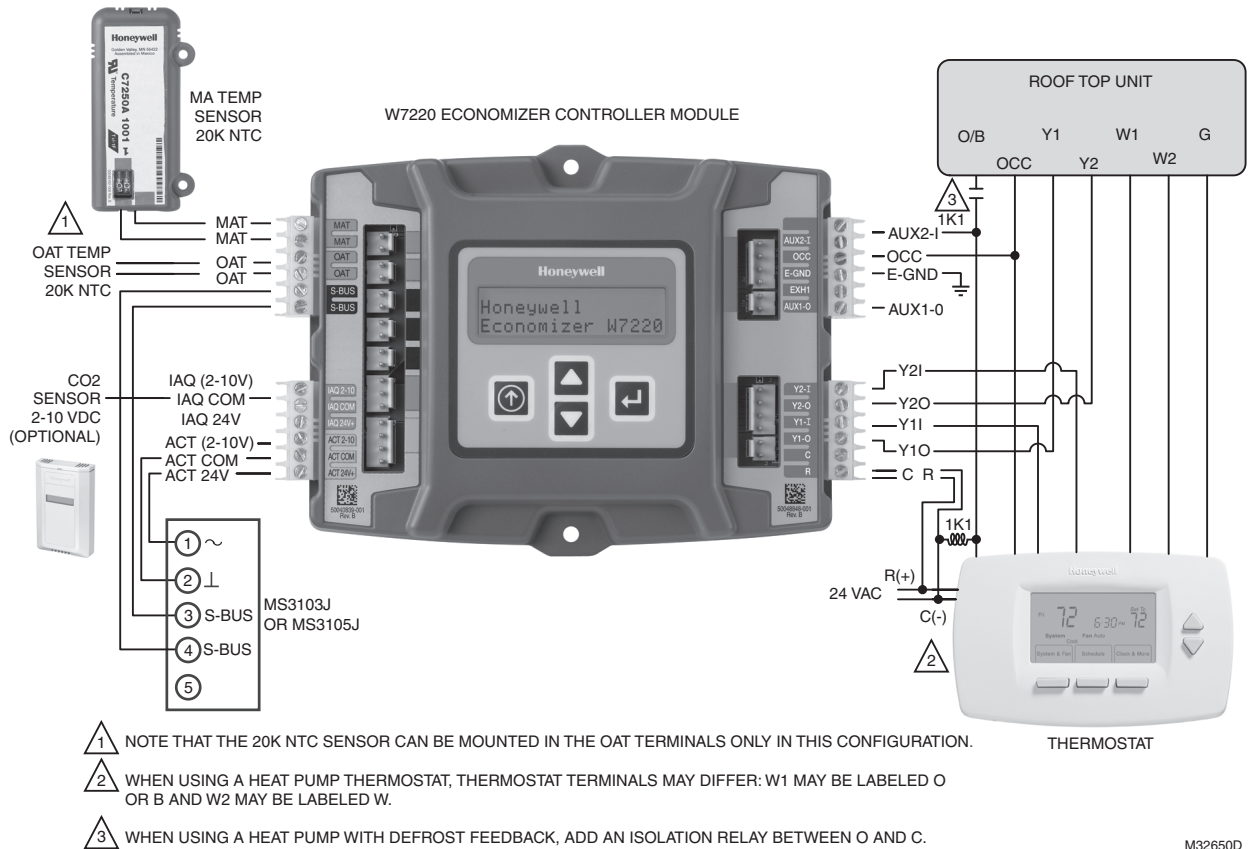
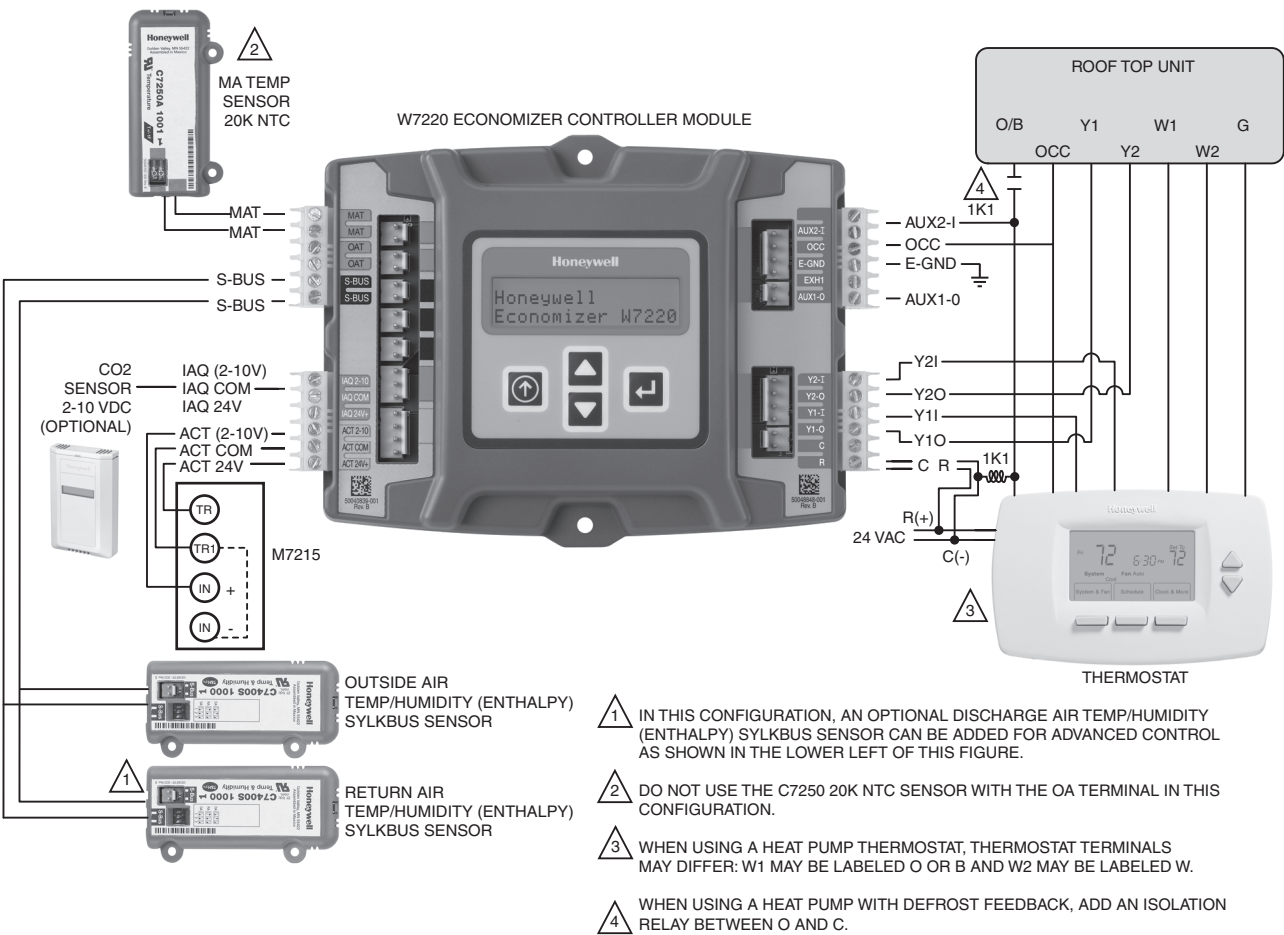
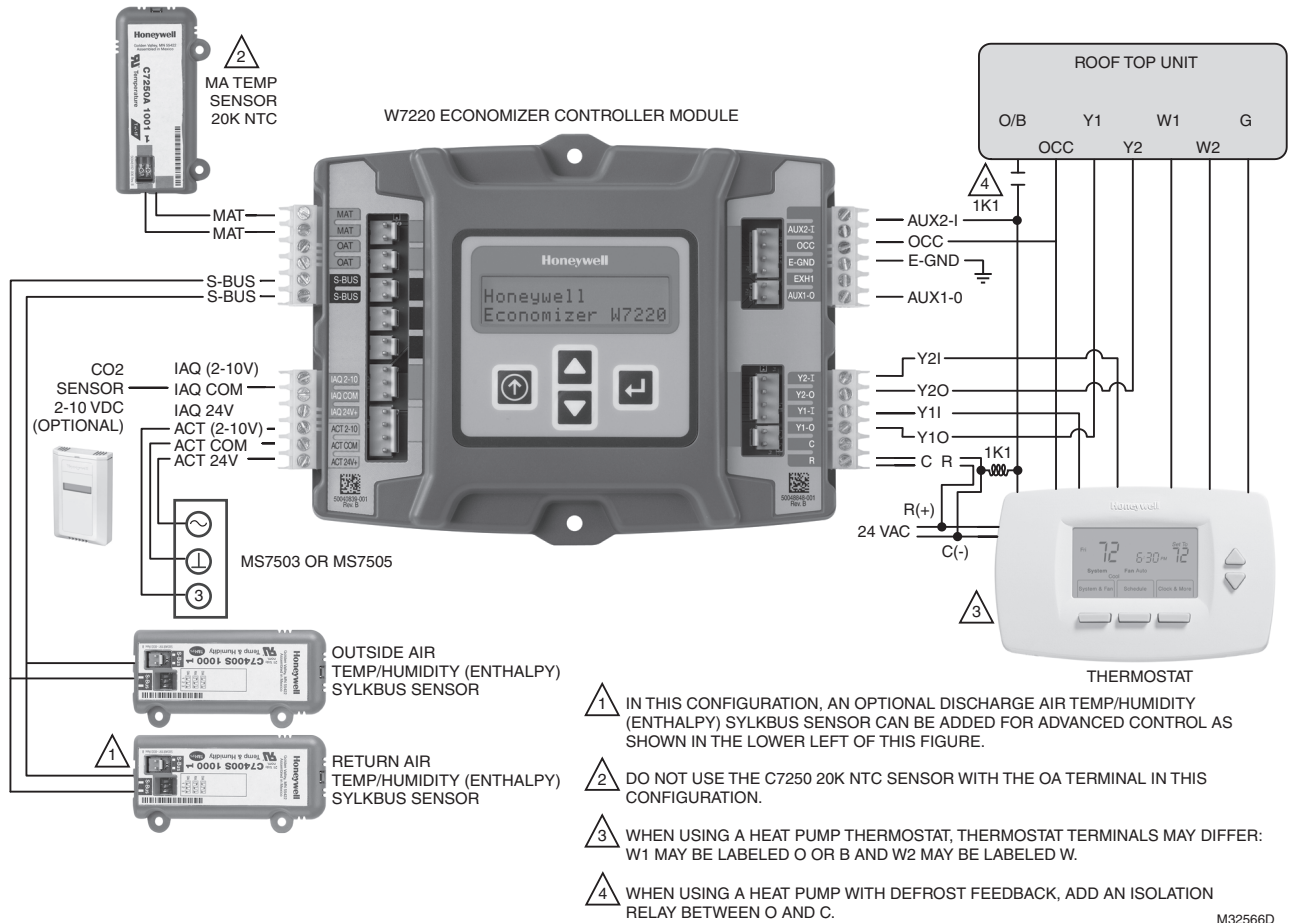


Fig. 3. Stand-alone dry-bulb Economizer configuration with Honeywell MS3103J or MS3105J communicating actuators.



M32565D

Fig. 4. Economizer with Sylk Bus sensors for enthalpy configuration with Honeywell M7215 black motor.



M32566D

Fig. 5. Economizer with Sylk Bus sensors for enthalpy configuration with a Honeywell MS7505 or MS7503 Direct Coupled Actuator.

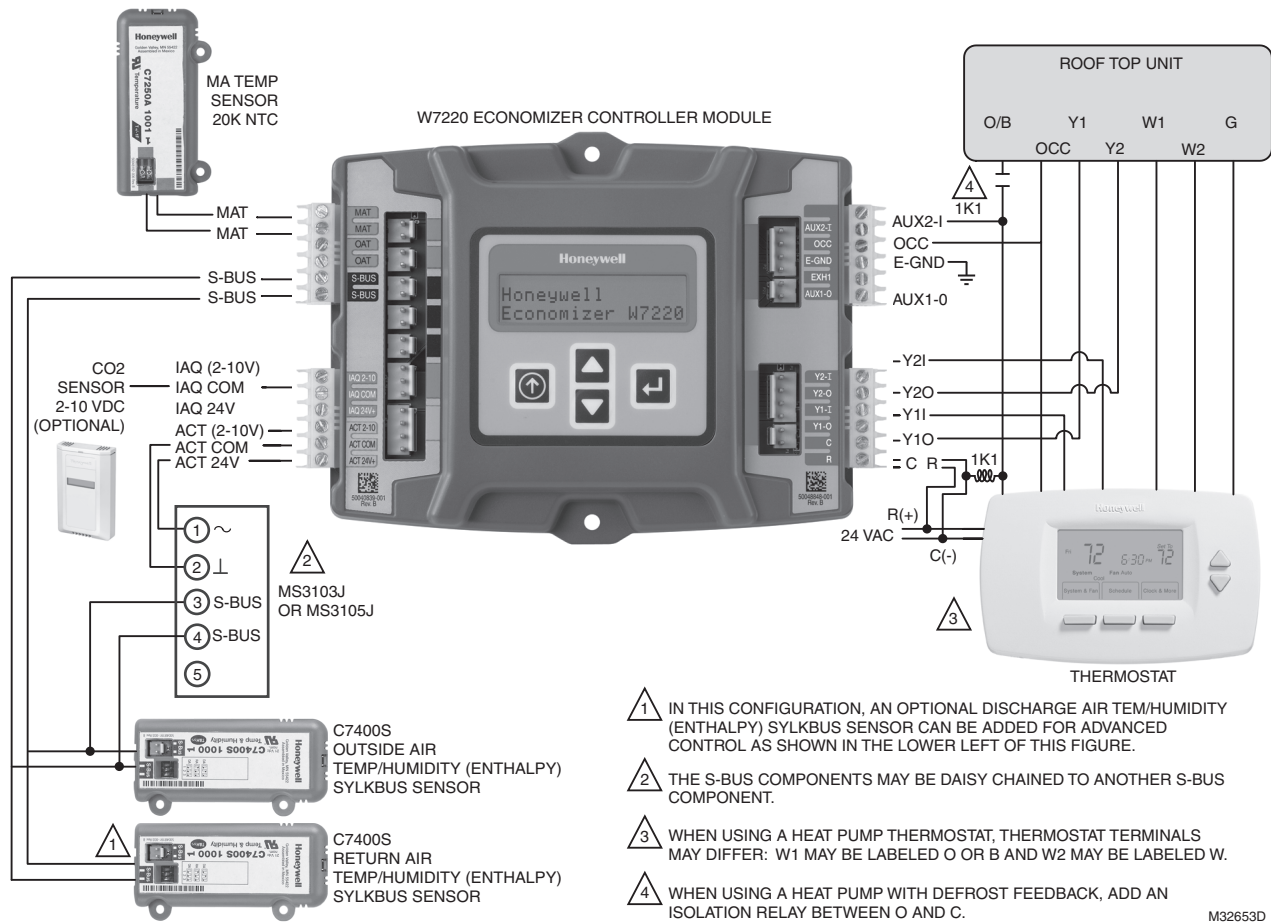


Fig. 6. Economizer with Sylk bus sensor for enthalpy configuration with Honeywell MS3103 or MS3105 communicating actuators.

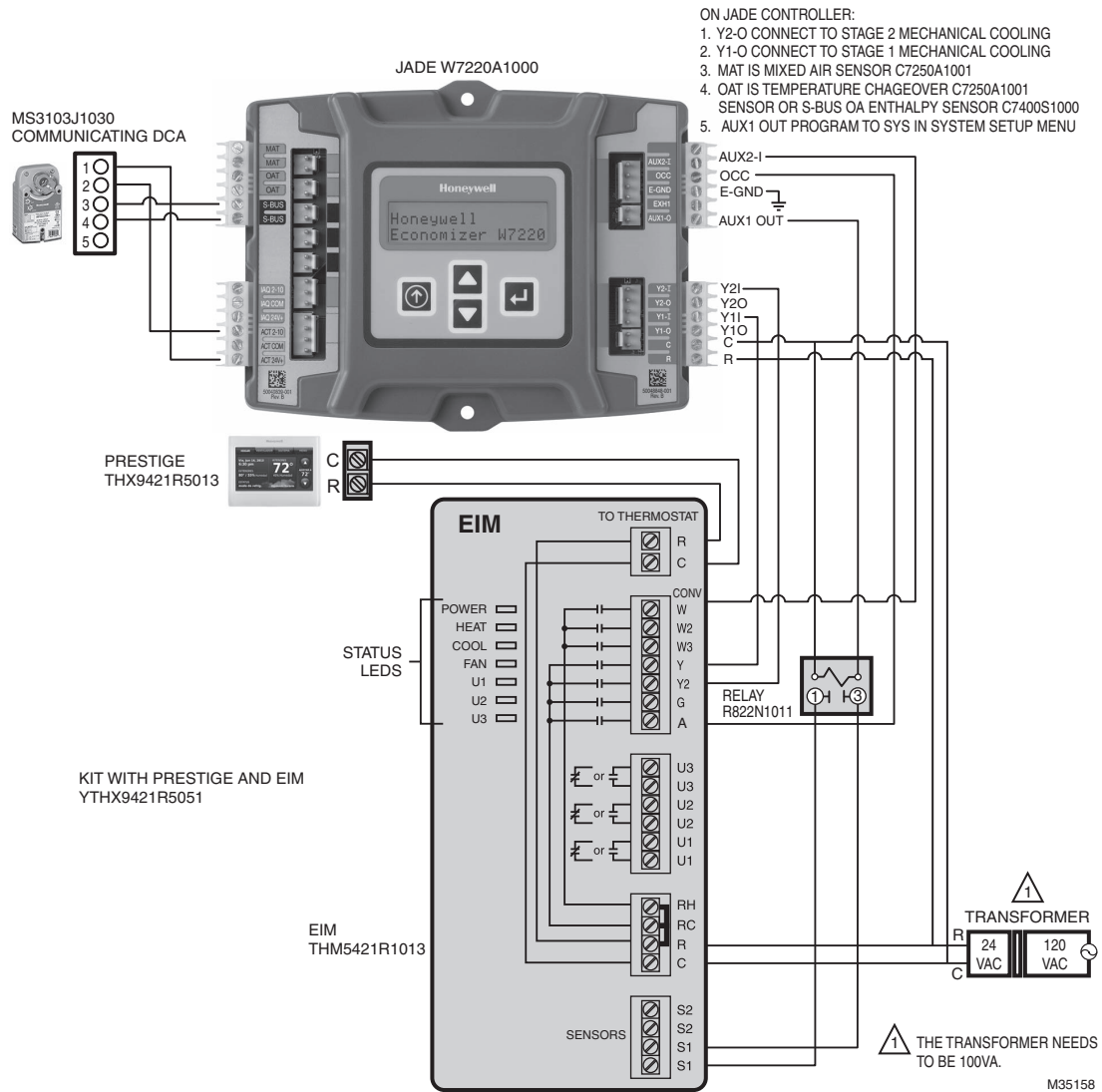
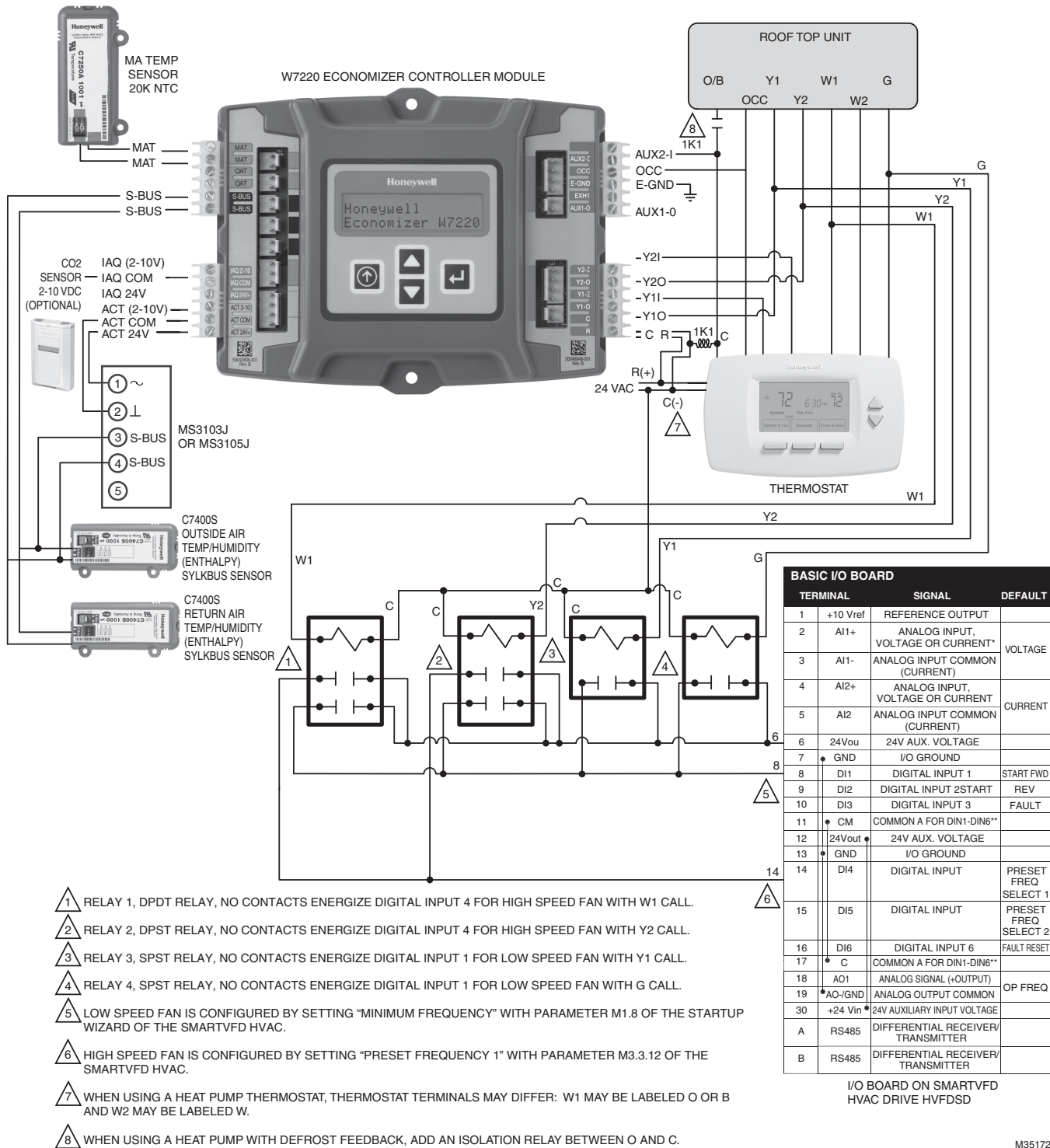
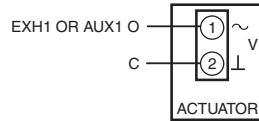


Fig. 7. Economizer with Honeywell MS3103 or MS3105 communicating actuator and thermostat EIM.



M35172

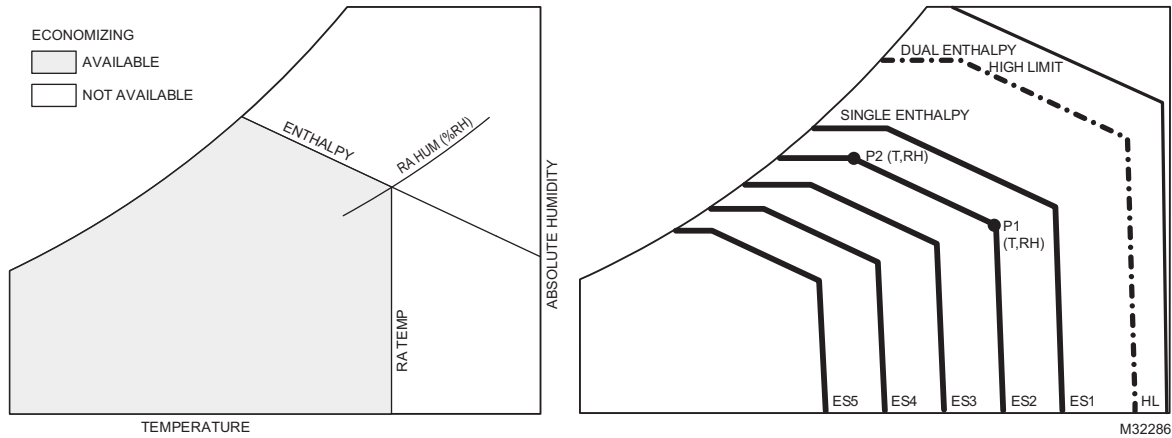
Fig. 8. Economizer with Sylk bus sensor for enthalpy configuration with Honeywell MS3103 or MS3105 communicating actuators, thermostat and VFD for Two-speed fan operation .



NOTE: ON/OFF ACTUATORS CAN BE USED ON THE EXH1 OR AUX1 O TERMINAL WITH GROUND TO THE C TERMINAL. WHEN PROGRAMMING THE EXH1 OR AUX1 O, THE % IS THE PERCENT OPEN POSITION OF THE OUTDOOR AIR DAMPER WHEN THE EXH1 OR AUX1 O TERMINAL IS ENERGIZED AND THE 2-POS DAMPER GOES OPEN. IF USING THE AUX1 O TERMINAL PROGRAM AUX1 O FOR EXH2.

M33409

Fig. 9. 2-position actuator.



M32286

Fig. 10. Single Enthalpy curves and boundaries.

Table 3. Single Enthalpy and Dual Enthalpy High Limit Curves.

Enthalpy Curve	Temp. Dry-Bulb (°F)	Temp. Dewpoint (°F)	Enthalpy (btu/lb/da)	Point P1		Point P2	
				Temp. °F	Humidity %RH	Temp. °F	Humidity %RH
ES1	80.0	60.0	28.0	80.0	36.8	66.3	80.1
ES2	75.0	57.0	26.0	75.0	39.6	63.3	80.0
ES3	70.0	54.0	24.0	70.0	42.3	59.7	81.4
ES4	65.0	51.0	22.0	65.0	44.8	55.7	84.2
ES5	60.0	48.0	20.0	60.0	46.9	51.3	88.5
HL	86.0	66.0	32.4	86.0	38.9	72.4	80.3

Power Up

After the module is mounted and wired, apply power.

Power Up Delay

Upon power up (or after a power outage or brownout), the W7220 controller module begins a 5 minute power up delay before enabling mechanical cooling.

Initial Menu Display

On initial start up, Honeywell displays on the first line and Economizer W7220 on the second line. After a brief pause, the software version appears on the first line and the second line will be blank. After another brief moment the screen will display W7220 on the first line and STATUS on the second line. Refer to table 3 for menu structure.

Power Loss (Outage or Brownout)

All setpoints and advanced settings are restored^a after any power loss or interruption.

NOTE: If power goes below 18 Vac, the W7220 controller module assumes a power loss and the 5 minute power up delay will become functional when power returns above 18 Vac.

Using the Keypad with Menus

To use the keypad when working with menus:

- Press the ▲ button to move to the previous menu.
- Press the ▼ button to move to the next menu.
- Press the ↵ button (Enter) to display the first item in the currently displayed menu.
- Press the ⬆ button (Menu up) to exit a menu's item and return to the list of menus.

Using the Keypad with Settings and Parameters

To use the keypad when working with Setpoints, System and Advanced Settings, Checkout tests, and Alarms:

- Navigate to the desired menu.
- Press the ↵ button (Enter) to display the first item in the currently displayed menu.
- Use the ▲ and ▼ buttons to scroll to the desired parameter.
- Press the ↵ button (Enter) to display the value of the currently displayed item.
- Press the ▲ button to increase (change) the displayed parameter value.^a

^a All settings are stored in non-volatile flash memory.

- Press the ▼ button to decrease (change) the displayed parameter value.^a
- Press the ↵ button to accept the displayed value and store it in non-volatile RAM.
- CHANGE STORED displays.
- Press the ↵ button (Enter) to return to the current menu parameter.
- Press the ⬆ button (MenuUp/Exit) to return to the previous menu.

^a When values are displayed, pressing and holding the ▲ or ▼ button causes the display to automatically increment.

Menu Structure

Table 4 illustrates the complete hierarchy of menus and parameters for the JADE™ Economizer system.

The Menus in display order are:

- STATUS
- SETPOINTS
- SYSTEM SETUP
- ADVANCED SETUP
- CHECKOUT
- ALARMS

IMPORTANT

Table 4 illustrates the complete hierarchy. Your menu parameters will be different depending on your configuration.

For example if you do not have a DCV (CO₂) sensor, then none of the DCV parameters appear and only MIN POS will display. If you have a CO₂ sensor, the DCV MIN and DCV MAX will appear AND if you have 2 speed fan DCV MIN (high and low speed) and DCV MAX (high and low speed will appear).

SETUP AND CONFIGURATION

Before being placed into service, the JADE™ Economizer module must be setup and configured for the installed system.

IMPORTANT

During setup, the Economizer module is live at all times.

The setup process uses a hierarchical menu structure that is easy to use. You press the ▲ and ▼ arrow buttons to move forward and backward through the menus and press the ↵ button to select and confirm setup item changes.

Table 4. Menu Structure^a.

Menu	Parameter	Parameter Default Value	Parameter Range and Increment ^b	Notes
STATUS	ECON AVAIL	NO	YES/NO	YES = economizing available; the system can use Outdoor Air for free cooling when required.
	ECONOMIZING	NO	YES/NO	YES = Outdoor Air being used for 1 st stage cooling.
	OCCUPIED	NO	YES/NO	YES = OCC signal received from space thermostat or unitary controller. YES = 24 Vac on terminal OCC No = 0 Vac on terminal OCC.
	HEAT PUMP	n/a ^c	COOL HEAT	Displays COOL or HEAT when system is set to heat pump (non-conventional)
	COOL Y1-IN	OFF	ON/OFF	Y1-I signal from space thermostat or unitary controller for cooling stage 1. ON = 24 Vac on term Y1-I OFF = 0 Vac on term Y1-I
	COOL Y1-OUT	OFF	ON/OFF	Cool Stage 1 Relay Output to mechanical cooling (Y1-OUT terminal).
	COOL Y2-IN	OFF	ON/OFF	Y2-I signal from space thermostat or unitary controller for second stage cooling. ON = 24 Vac on term Y2-I OFF = 0 Vac on term Y2-I
	COOL Y2-OUT	OFF	ON/OFF	Cool Stage 2 Relay Output to mechanical cooling (Y2-OUT terminal).
	MA TEMP	-- °F	-40 to 150 °F	Displays value of measured mixed air from MAT sensor. Displays -- if not connected, short, or out-of-range.
	DA TEMP	-- °F	-40 to 150 °F	Displays when Discharge Air Sylk Bus sensor is connected and displays measured discharge air temperature. Displays -- °F if sensor sends invalid value, if not connected, short or out-of-range.
	OA TEMP	-- °F	-40 to 140 °F	Displays measured value of outdoor air temperature. Displays -- °F if sensor sends invalid value, if not connected, short or out-of-range.
	OA HUM	-- %	0 to 100%	Displays measured value of outdoor humidity from OA Sylkbus sensor. Displays --% if not connected, short, or out-of-range.
	RA TEMP	-- °F	0 to 140 °F	Displays measured value of return air temperature from RA Sylkbus sensor. Displays -- °F if sensor sends invalid value, if not connected, short or out-of-range.
	RA HUM	-- %	0 to 100%	Displays measured value of return air humidity from RA Sylkbus sensor. Displays --% if sensor sends invalid value, if not connected, short or out-of-range.
	IN CO2	-- ppm	0 to 2000 ppm	Displays value of measured CO2 from CO2 sensor. Invalid if not connected, short or out-of-range. May be adjusted in Advanced menu by Zero offset and Span.
	DCV STATUS	n/a	ON/OFF	Displays ON if above setpoint and OFF if below setpoint, and ONLY if a CO2 sensor is connected.
	DAMPER OUT	2.0V	2.0 to 10.0 V	Displays output voltage or position to the damper actuator. ^e
	ACT POS	n/a	0 to 100%	Displays actual position of actuator.
	ACT COUNT	n/a	1 to 65,535	Displays number of times actuator has cycled. 1 Cycle equals the sum of 180° of movement in any direction.
	ACTUATOR	n/a	OK/Alarm (on Alarm menu)	Displays Error if voltage or torque is below actuator range
	EXH1 OUT	OFF	ON/OFF	Output of EXH1 terminal. Displays ON when damper position reaches programmed percentage setpoint. ON = 24 Vac Output; OFF = No Output.

Table 4. Menu Structure^a. (Continued)

Menu	Parameter	Parameter Default Value	Parameter Range and Increment ^b	Notes
STATUS CONTINUED	EXH2 OUT	OFF	ON/OFF	Output of AUX1 0 terminal, if programmed for EXH2. Displays ON when damper position reaches programmed percentage setpoint ON = 24 Vac Output, OFF = No Output; displays only if AUX1 0 = EXH2
	ERV	OFF	ON/OFF	Output of AUX1 0 terminal, if programmed for ERV. ON = 24 Vac Output, OFF = No Output; displays only if AUX1 0 = ERV
	MECH COOL ON or HEAT STAGES ON	0	0, 1, or 2	Displays number of mechanical cooling stages that are active. Displays the stage of heat pump heating that is active.
	FAN SPEED	2 speed	LOW or HIGH	Displays speed of fan if JADE is programmed for 2-speed fan.
	W (HEAT IN)	n/a	ON/OFF	Displays status of heat if JADE is programmed for 2-speed fan.

Table 4. Menu Structure^a. (Continued)

Menu	Parameter	Parameter Default Value	Parameter Range and Increment ^b	Notes
SETPOINTS	MAT SET	53°F	38 to 70 °F; increment by 1	The economizer will modulate the OA damper to maintain the mixed air temperature at the setpoint.
	LOW T LOCK	32°F	-45 to 80 °F; increment by 1	Setpoint determines outdoor temperature when the mechanical cooling cannot be turned on. Commonly referred to as the Compressor lockout. At or below the setpoint the Y1-O and Y2-O will not be energized on the controller.
	DRYBLB SET	63°F	48 to 80 °F; increment by 1	<i>Dry bulb setpoint will only appear if using dry bulb change over.</i> Setpoint determines where the economizer will assume outdoor air temperature is good for free cooling; e.g.; at 63 °F setpoint unit will economize at 62 °F and below and not economize at 64 °F and above. There is a 2 °F deadband.
	DRYBLB DIF (Available firmware 1.15, June 2018, and later.)	0°F	0 to 6°F; increment by 2	Drybulb Differential will only show if using dual drybulb, i.e., when an outdoor air temperature sensor C7250 is attached to OAT terminals, and a C7400S sensor is wired to S-Bus and configured for RAT (return air). Free cooling will be assumed whenever OA Temp is below RAT minus this drybulb differential setting.
	ENTH CURVE	ES3	ES1, ES2, ES3, ES4, or ES5	<i>ES curve will only appear if using enthalpy changeover.</i> Enthalpy boundary “curves” for economizing using single enthalpy. See page 11 for description of enthalpy curves.
	DCV SET	1100ppm	500 to 2000 ppm increment by 100	Displays ONLY if a CO2 sensor is connected. Setpoint for Demand Control Ventilation of space. Above the setpoint, the OA dampers will modulate open to bring in additional OA to maintain a space ppm level below the setpoint.
	MIN POS	2.8 V	2 to 10 Vdc	Displays ONLY if a CO2 sensor is NOT connected. With 2-speed fan units MIN POS L (low speed fan) and MIN POS H (high speed fan) settings are required. Default for MIN POS L is 3.2V and MIN POS H is 2.8V.
	VENTMAX	2.8 V	2 to 10 Vdc	Displays only if a CO2 sensor is connected. Used for Vbz (ventilation max cfm) setpoint. VENTMAX is the same setting as MIN POS would be if you did not have the CO2 sensor.
			100 to 9990 cfm increment by 10	If OA, MA RA and CO2 sensors are connected and DCV CAL ENABLE is set to AUTO mode, the OA dampers are controlled by CFM and displays from 100 to 9990 cfm.
			2 to 10 Vdc	With 2-speed fan units VENTMAX L (low speed fan) and VENTMAX H (high speed fan) settings are required. Default for VENTMAX L is 3.2V and VENTMAX H is 2.8V.
	VENTMIN	2.25 V	2 to 10 Vdc	Displays only if CO2 sensor is connected. Used for Va (ventilation min cfm) setpoint. This is the ventilation requirement for less than maximum occupancy of the space.
			100 to 9990 cfm increment by 10	If OA, MA RA and CO2 sensors are connected and DCV CAL ENABLE is set to AUTO mode, the OA dampers are controlled by CFM and displays from 100 to 9990 cfm.
			2 to 10 Vdc	With 2-speed fan units VENTMIN L (low speed fan) and VENTMIN H (high speed fan) settings are required. Default for VENTMIN L is 2.5V and VENTMIN H is 2.25V.

Table 4. Menu Structure^a. (Continued)

Menu	Parameter	Parameter Default Value	Parameter Range and Increment ^b	Notes
SETPOINTS CONTINUED	ERV OAT SP ^d	32°F	0 to 50 °F; increment by 1	Only when AUX1 0 = ERV
	EXH1 SET	50%	0 to 100%; increment by 1	Setpoint for OA damper position when exhaust fan 1 is powered by the economizer. With 2-speed fan units Exh1 L (low speed fan) and Exh1 H (high speed fan) settings are required. Default for Exh1 L is 65% and Exh1 H is 50%
	EXH2 SET	75%	0 to 100%; increment by 1	Setpoint for OA damper position when exhaust fan 2 is powered by the economizer. Only used when AUX1 0 is set to EXH2. With 2-speed fan units Exh2 L (low speed fan) and Exh2 H (high speed fan) settings are required. Default for Exh2 L is 80% and Exh2 H is 75%
SYSTEM SETUP	INSTALL	01/01/11		Display order = MM/DD/YY Setting order = DD, MM, then YY.
	UNITS DEG	°F	°F or °C	Sets economizer controller in degrees Fahrenheit or Celsius.
	EQUIPMENT	CONV	CONV HP	CONV = conventional. HP O/B = Enables Heat Pump mode. Use AUX2 I for Heat Pump input from thermostat or controller (O or B terminal).
	AUX2 IN	Heat (W1) ^g Note: If field factory default is available, AUX2 IN will default back to Shutdown (SD).	Shutdown (SD) Heat (W1) HP(O) HP(B)	In CONV mode: SD = Enables configuration of shutdown (default); W = Informs controller that system is in heating mode. NOTE: If using 2-speed fan mode, you must program CONV mode for W. Shutdown is not available in the two-speed fan mode. In HP O/B mode: HP(O) = energize heat pump on Cool (default); HP(B) = energize heat pump on Heat.
	FAN SPEED	2 speed ^h Note: If field factory default is available, FAN SPEED will default back to 1-speed.	1 speed/ 2 speed	Sets economizer controller for operation of 1 speed or 2 speed supply fan. The controller does not control the fan but positions the OA and RA dampers to the heating or cooling mode settings. NOTE: 2-speed fan option also needs Heat (W1) programmed in AUX 2 In.
	FAN CFM	5000cfm	100 to 50000 cfm; increment by 100	This is the capacity of the fan. The value is found on the label from the RTU manufacturer. The cfm of the fan is only used with DCV CAL ENABLE AUO
	AUX1 OUT	NONE	NONE ERV EXH2 SYS	- NONE = not configured (output is not used) - ERV= Energy Recovery Ventilator^d - EXH2 = second damper position 24 Vac out for second exhaust fan. - SYS = use output as an alarm signal
	OCC	INPUT	INPUT or ALWAYS	When using a setback thermostat with occupancy out (24 Vac), the 24 Vac is input "INPUT" to the OCC terminal. If no occupancy output from the thermostat then change program to "ALWAYS" OR add a jumper from terminal R to OCC terminal.

Table 4. Menu Structure^a. (Continued)

Menu	Parameter	Parameter Default Value	Parameter Range and Increment ^b	Notes
ADVANCED SETUP	MA LO SET	45 °F	35 to 65 °F; increment by 1°	Temp to activate Freeze Protection (close damper or modulate to MIN POS if Mixed air temperature falls below set value)
	FREEZE POS	CLO	CLO MIN	Damper position when freeze protection is active (closed or MIN POS).
	CO2 ZERO	0ppm	0 to 500 ppm; increment by 10	CO2 ppm level to match CO2 sensor start level.
	CO2 SPAN	2000ppm	1000 to 3000 ppm; increment by 50	CO2 ppm span to match CO2 sensor; e.g.; 500-1500 sensor output would be 500 CO2 zero and 1000 CO2 span.
	STG3 DLY	2.0h	0 min, 5 min, 15 min, then 15 min intervals. Up to 4h or OFF	Delay after stage 2 (Y2 IN) for cool has been active. Turns on 2nd stage of mechanical cooling when economizer is 1st stage call and mechanical cooling is 2nd stage call. Allows three stages of cooling, 1 economizer and 2 mechanical. OFF = no Stage 3 cooling.
	SD DMPR POS	CLO	CLO OPN	Indicates shutdown signal when JADE is programmed for SHUTDOWN. When controller receives 24 Vac input on the AUX2 IN terminal in conventional mode, the OA damper will open if programmed for OPN and the OA damper will close if programmed for CLO. All other controls, e.g., Y1-O, Y2-O, EXH1, etc. will shut off.
	DA LO ALM	45 °F	NONE 35°F to 65°F in 5°F increments	Used for alarm for when the DA air temperature is too low. Set lower range of alarm, below this temperature the alarm will show on the display. Must have a DA temperature sensor (C7400S programmed for DA) in the discharge air.
	DA HI ALM	80 °F	NONE 70 °F to 180 °F in 5° F increments	Used for alarm for when the DA air temperature is too high. Set high range of alarm, above this temperature the alarm will show on the display. Must have a DA temperature sensor (C7400S programmed for DA) in the discharge air.
	DCVCAL ENA	MAN	MAN (manual) AUTO	Turns on the DCV automatic control of the dampers. Resets ventilation based on the RA, OA and MA sensor conditions. Requires all sensors (RA, OA, MA and CO2). This operation is not operable with a 2-speed fan unit.
	MAT T CAL	0.0 F°	+/-2.5F°	Allows for the operator to adjust for an out of calibration temperature sensor
	OAS T CAL	0.0F°	+/-2.5F°	Allows for the operator to adjust for an out of calibration temperature sensor
	OAS H CAL	0% RH	+/-10% RH	Allows for the operator to adjust for an out of calibration humidity sensor
	RA T CAL	0.0F°	+/-2.5F°	Allows for the operator to adjust for an out of calibration temperature sensor
	RA H CAL	0% RH	+/-10% RH	Allows for the operator to adjust for an out of calibration humidity sensor
	DA T CAL	0.0 F°	+/-2.5F°	Allows for the operator to adjust for an out of calibration temperature sensor
	2SP FAN DELAY	5 Minutes	0 to 20 minutes in 1 minute increments.	When in economizing mode this is the delay for the high speed fan to try to satisfy the call for second stage cooling before the first stage mechanical cooling is enabled.

Table 4. Menu Structure^a. (Continued)

Menu	Parameter	Parameter Default Value	Parameter Range and Increment ^b	Notes
CHECKOUT ^f	DAMPER MINIMUM POSITION	n/a	n/a	The checkout for the damper minimum positions is based on the system. See Table 5.
	DAMPER OPEN	n/a	n/a	Positions damper to the full open position. Exhaust fan contacts enable during the DAMPER OPEN test. Make sure you pause in this mode to allow for exhaust contacts to energize due to the delay in the system.
	DAMPER CLOSE	n/a	n/a	Positions damper to the fully closed position.
	CONNECT Y1-O	n/a	n/a	Closes the Y1-O relay (Y1-O). See CAUTION on page 24
	CONNECT Y2-O	n/a	n/a	Closes the Y2-O relay (Y2-O). See CAUTION on page 24
	CONNECT AUX1-O	n/a	n/a	Energizes the AUX1-O output.
	CONNECT EXH1	n/a	n/a	Closes the power exhaust fan 1 relay (EXH1)
ALARMS(#)				Alarms display only when they are active. The menu title "ALARMS (#)" includes the number of active alarms in parenthesis (). When using SYLK bus sensors, "SYLK" will appear on the screen, and when using 20k OA temperature sensors, "SENS T" will appear on the screen.
	MA T SENS ERR	n/a	n/a	Mixed air sensor has failed or become disconnected - check wiring then replace sensor if the alarm continues
	CO2 SENS ERR	n/a	n/a	CO2 sensor has failed, gone out of range or become disconnected - check wiring then replace sensor if the alarm continues
	OA SYLK T ERR	n/a	n/a	Outdoor air enthalpy sensor has failed or become disconnected - check wiring then replace sensor if the alarm continues
	OA SYLK H ERR	n/a	n/a	
	RA SYLK T ERR	n/a	n/a	Return air enthalpy sensor has failed or become disconnected - check wiring then replace sensor if the alarm continues
	RA SYLK H ERR	n/a	n/a	
	DA SYLK T ERR	n/a	n/a	Discharge air sensor has failed or become disconnected - check wiring then replace sensor if the alarm continues
	OA SENS T ERR	n/a	n/a	Outdoor air temperature sensor has failed or become disconnected - check wiring then replace sensor if the alarm continues
	ACT ERROR	n/a	n/a	Actuator has failed or become disconnected - check for stall, over voltage, under voltage and actuator count. Replace actuator if damper is moveable and supply voltage is between 21.6 V and 26.4 V. Check actuator count on STATUS menu.
	FREEZE ALARM	n/a	n/a	Check if outdoor temperature is below the LOW Temp Lockout on setpoint menu. Check if Mixed air temperature on STATUS menu is below the Lo Setpoint on Advanced setup menu. When conditions are back in normal range then the alarm will go away.
	SHUTDOWN ACTIVE	n/a	n/a	AUX2 IN is programmed for SHUTDOWN and 24 V has been applied to AUX2 IN terminal
	DMP CAL RUNNING	n/a	n/a	If DCV Auto enable has been programmed, when the JADE is completing a calibration on the dampers, this alarm will display. Wait until the calibration is completed and the alarm will go away. Must have OA, MA and RA sensors for DCV calibration; set up is in the Advanced setup menu
	DA SENS ALM	n/a	n/a	Discharge air temperature is out of the range set in the ADVANCED SETUP Menu. Check the temperature of the discharge air.
ALARMS(#) CONTINUED	SYS ALARM	n/a	n/a	When AUX1-O is set to SYS and there is any alarm (e.g., failed sensors, etc.), the AUX1-O terminal has 24 Vac out.
	ACT UNDER V	n/a	n/a	Voltage received by Actuator is below expected range
	ACT OVER V	n/a	n/a	Voltage received by Actuator is above expected range
	ACT STALLED	n/a	n/a	Actuator stopped before achieving commanded position

Table 4. Menu Structure^a. (Continued)

Menu	Parameter	Parameter Default Value	Parameter Range and Increment ^b	Notes
FEATURES ADJUSTABLE ONLY BY USE OF THE W7220 PC MOD TOOL				
	ACT STALL ALARM SUPPRESSION	Disabled	Enabled or Disabled	If enabled this feature allows the operator to suppress the stall alarm in a specific range of the actuator stroke.

^a Table 4 illustrates the complete hierarchy. Your menu parameters may be different depending on your configuration. For example if you do not have a DCV (CO₂) sensor, then none of the DCV parameters appear.

^b When values are displayed, pressing and holding the ▲ or ▼ button causes the display to automatically increment.

^c n/a = not applicable

^d ERV Operation: When in Cooling mode AND the conditions are NOT OK for economizing - the ERV terminal will be energized. In the Heating mode the ERV terminal will be energized when the OA is below the ERV OAT setpoint in the setpoint menu.

^e When used with Honeywell communicating actuator the damper out is reported in XX.X% open versus XX.X Vdc.

^f After 10 minutes without a command or mode change, the controller will change to normal operation.

^g Default for AUX2 IN changed from SD to W1 with 1.15 firmware (June 2018).

^h Default for FAN SPEED changed from 1-speed to 2-speed with 1.15 firmware (June 2018).

Table 5. Damper minimum position settings and readings on checkout menu.

Fan Speed	Demand Control Ventilation (CO ₂ Sensor)	Setpoints	Checkout
1	NO	MIN POS	VMAX-HS
1	NO	N/A	N/A
2	NO	MIN POS H	VMAX-HS
2	NO	MIN POS L	VMAX-LS
1	YES	VENT MIN	VMIN-HS
1	YES	VENT MAX	VMAX-HS
2	YES	VENT MIN H	VMIN-HS
2	YES	VENT MAX H	VMAX-LS
2	YES	VENT MINL	N/A
2	YES	VENT MAX L	N/A

SEQUENCE OF OPERATION

Table 6. Dry Bulb Operation No DCV (CO2 sensor) - 1 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
None	No	Off	Off	High	0-v/Off	0-v/Off	MIN POS	Closed
		On	Off	High	24-v/On	0-v/Off	MIN POS	Closed
		On	On	High	24-v/On	24-v/On	MIN POS	Closed
None	Yes	Off	Off	High	0-v/Off	0-v/Off	MIN POS	Closed
		On	Off	High	0-v/Off	0-v/Off	MIN POS to Full-Open	Closed to Full-Open
		On	On	High	24-v/On	0-v/Off ^a	MIN POS to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

Table 7. Dry Bulb Operation With DCV (CO2 sensor) - 1 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
Below CO2 set	No	Off	Off	High	0-v/Off	0-v/Off	VENTMIN	Closed
		On	Off	High	24-v/On	0-v/Off	VENTMIN	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN	Closed
	Yes	Off	Off	High	0-v/Off	0-v/Off	VENTMIN	Closed
		On	Off	High	0-v/Off	0-v/Off	VENTMIN to Full-Open	Closed to Full-Open
		On	On	High	24-v/On	0-v/Off ^a	VENTMIN to Full-Open	Closed to Full-Open
Above CO2 set	No	Off	Off	High	0-v/Off	0-v/Off	VENTMIN to VENTMAX	Closed
		On	Off	High	24-v/On	0-v/Off	VENTMIN to VENTMAX	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN to VENTMAX	Closed
	Yes	Off	Off	High	0-v/Off	0-v/Off	VENTMIN to VENTMAX	Closed
		On	Off	High	0-v/Off	0-v/Off	VENTMIN to Full-Open	Closed to Full-Open
		On	On	High	24-v/On	0-v/Off ^a	VENTMIN to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

Table 8. Enthalpy Operation No DCV (CO2 sensor) - 1 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
None	No	Off	Off	High	0-v/Off	0-v/Off	MIN POS	Closed
		On	Off	High	24-v/On	0-v/Off	MIN POS	Closed
		On	On	High	24-v/On	24-v/On	MIN POS	Closed
None	Yes	Off	Off	High	0-v/Off	0-v/Off	MIN POS	Closed
		On	Off	High	0-v/Off	0-v/Off	MIN POS to Full-Open	Closed to Full-Open
		On	On	High	24-v/On	0-v/Off ^a	MIN POS to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

Table 9. Enthalpy Operation With DCV (CO2 sensor) - 1 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
Below set	No	Off	Off	High	0-v/Off	0-v/Off	VENTMIN	Closed
		On	Off	High	24-v/On	0-v/Off	VENTMIN	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN	Closed
	Yes	Off	Off	High	0-v/Off	0-v/Off	VENTMIN	Closed
		On	Off	High	0-v/Off	0-v/Off	VENTMIN to Full-Open	Closed to Full-Open
		On	On	High	24-v/On	0-v/Off ^a	VENTMIN to Full-Open	Closed to Full-Open
Above set	No	Off	Off	High	0-v/Off	0-v/Off	VENTMIN to VENTMAX	Closed
		On	Off	High	24-v/On	0-v/Off	VENTMIN to VENTMAX	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN to VENTMAX	Closed
	Yes	Off	Off	High	0-v/Off	0-v/Off	VENTMIN to VENTMAX	Closed
		On	Off	High	0-v/Off	0-v/Off	VENTMIN to Full-Open	Closed to Full-Open
		On	On	High	DELAY (b) 24-v/On	0-v/Off ^a	VENTMIN to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

Table 10. Dry Bulb Operation No DCV (CO2 sensor) - 2 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
None	No	Off	Off	Low	0-v/Off	0-v/Off	MIN POS L	Closed
		On	Off	Low	24-v/On	0-v/Off	MIN POS L	Closed
		On	On	High	24-v/On	24-v/On	MIN POS H	Closed
None	Yes	Off	Off	Low	0-v/Off	0-v/Off	MIN POS L	Closed
		On	Off	Low	0-v/Off	0-v/Off	MIN POS L to Full-Open	Closed to Full-Open
		On	On	High	DELAY ^b 24-v/On	0-v/Off ^a	MIN POS H to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

^b With 2SP FAN DELAY (Advanced Setup Menu) when in the economizing mode there is a delay for the high speed fan to try to satisfy the call for second stage cooling by turning on the fan to high and opening the OA damper 100% before the first stage mechanical cooling is enabled.

Table 11. Dry Bulb Operation With DCV (CO2 sensor) - 2 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
Below set	No	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L	Closed
		On	Off	Low	24-v/On	0-v/Off	VENTMIN L	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN H	Closed
	Yes	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L	Closed
		On	Off	Low	0-v/Off	0-v/Off	VENTMIN L to Full-Open	Closed to Full-Open
		On	On	High	24-v/On	0-v/Off ^a	VENTMIN H to Full-Open	Closed to Full-Open
Above set	No	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L to VENTMAX L	Closed
		On	Off	Low	24-v/On	0-v/Off	VENTMIN L to VENTMAX L	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN H to VENTMAX H	Closed
	Yes	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L to VENTMAX L	Closed
		On	Off	Low	0-v/Off	0-v/Off	VENTMIN L to Full-Open	Closed to Full-Open
		On	On	High	DELAY ^b 24-v/On	0-v/Off ^a	VENTMIN H to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

^b With 2SP FAN DELAY (Advanced Setup Menu) when in the economizing mode there is a delay for the high speed fan to try to satisfy the call for second stage cooling by turning on the fan to high and opening the OA damper 100% before the first stage mechanical cooling is enabled.

Table 12. Enthalpy Operation No DCV (CO2 sensor) - 2 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
NO CO2 SENSOR	No	Off	Off	Low	0-v/Off	0-v/Off	MIN POS L	Closed
		On	Off	Low	24-v/On	0-v/Off	MIN POS L	Closed
		On	On	High	24-v/On	24-v/On	MIN POS H	Closed
	Yes	Off	Off	Low	0-v/Off	0-v/Off	MIN POS L	Closed
		On	Off	Low	0-v/Off	0-v/Off	MIN POS L to Full-Open	Closed to Full-Open
		On	On	High	DELAY ^b 24-v/On	0-v/Off ^a	MIN POS H to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

^b With 2SP FAN DELAY (Advanced Setup Menu) when in the economizing mode there is a delay for the high speed fan to try to satisfy the call for second stage cooling by turning on the fan to high and opening the OA damper 100% before the first stage mechanical cooling is enabled.

Table 13. Enthalpy Operation With DCV (CO2 sensor) - 2 Speed Fan.

DCV	OA Good to economize?	Y1-I	Y2-I	FAN SPD	Y1-O	Y2-O	Occupied	Unoccupied
Below set	No	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L	Closed
		On	Off	Low	24-v/On	0-v/Off	VENTMIN L	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN H	Closed
	Yes	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L	Closed
		On	Off	Low	0-v/Off	0-v/Off	VENTMIN L to Full-Open	Closed to Full-Open
		On	On	High	24-v/On	0-v/Off ^a	VENTMIN H to Full-Open	Closed to Full-Open
Above set	No	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L to VENTMAX L	Closed
		On	Off	Low	24-v/On	0-v/Off	VENTMIN L to VENTMAX L	Closed
		On	On	High	24-v/On	24-v/On	VENTMIN H to VENTMAX H	Closed
	Yes	Off	Off	Low	0-v/Off	0-v/Off	VENTMIN L to VENTMAX L	Closed
		On	Off	Low	0-v/Off	0-v/Off	VENTMIN L to Full-Open	Closed to Full-Open
		On	On	High	DELAY ^b 24-v/On	0-v/Off ^a	VENTMIN H to Full-Open	Closed to Full-Open

^a With stage 3 delay (STG3 DLY) in Advanced setup menu can turn on 2nd stage of mechanical cooling Y2 -O after the delay if the call for Y1-I and Y2-I have not been satisfied.

^b With 2SP FAN DELAY (Advanced Setup Menu) when in the economizing mode there is a delay for the high speed fan to try to satisfy the call for second stage cooling by turning on the fan to high and opening the OA damper 100% before the first stage mechanical cooling is enabled.

TROUBLESHOOTING

Alarms

The Economizer module provides alarm messages that display on the 2-line LCD.

NOTE: JADE will be in the “set up” mode for the first 60 minutes after powered. If a sensor for OA air or Sylkbus device (sensor, actuator) is disconnected during the set up mode, the JADE will not alarm that failure. The MA sensor is a system “critical” sensor; if the MA sensor is removed during the set up mode, the JADE will alarm. After 60 minutes the JADE controller will change to operation mode and all components removed or failed will alarm in the operation mode.

You can also navigate to the Alarms menu at any time.

Clearing Alarms

Once the alarm has been identified and the cause has been removed (e.g. replaced faulty sensor), the alarm can be cleared from the display.

To clear an alarm, perform the following:

1. Navigate to the desired alarm.
2. Press the  button.
3. ERASE? displays.
4. Press the  button.
5. ALARM ERASED displays.
6. Press the  button (MenuUp/Exit) to complete the action and return to the previous menu.

NOTE: If an alarm still exists after you clear it, it re-displays within 5 seconds.

Checkout Tests

Use the Checkout menu (Table 4) to test the damper operation and any configured outputs. Only items that are configured are shown in the Checkout menu.

NOTE: See “Interface Overview” on page 18 of 63-2700 for information about menu navigation and use of the keypad.

To perform a Checkout test:

1. Scroll to the desired test in the Checkout menu using the  and  buttons.
2. Press the  button to select the item.
3. RUN? appears on the display.
4. Press the  button to start the test.
5. The unit pauses and then displays IN PROGRESS.
6. When all parameters have been tested, press the  button (Menu up) to end the test (e.g. turn off the relay).

The checkout tests can all be performed at the time of installation or any time during the operation of the system as a test that the system is operable.



CAUTION

Equipment damage may result.

Be sure to allow enough time for compressor startup and shutdown between checkout tests so that you do not short-cycle the compressors

By using this Honeywell literature, you agree that Honeywell will have no liability for any damages arising out of your use or modification to, the literature. You will defend and indemnify Honeywell, its affiliates and subsidiaries, from and against any liability, cost, or damages, including attorneys’ fees, arising out of, or resulting from, any modification to the literature by you.

Home and Building Technologies

In the U.S.:

Honeywell

715 Peachtree Street NE

Atlanta, GA 30308

customer.honeywell.com

Honeywell

® U.S. Registered Trademark
© 2018 Honeywell International Inc.
62-0331-15 M.S. Rev. 05-18
Printed in United States